

Investigating Hebbian Alternatives to Dense Associative Memory

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★ Motivations

- (a) Hopfield networks (J. J. Hopfield, 1984; J. Hopfield, 1982) are a fruitful abstraction that captures associative memory behaviour, and has productive generalizations (Hoover et al., 2023; Krotov & Hopfield, 2016; Ramsauer et al., 2021)
- (b) But these generalizations trade Hebbian learning rules for back-propagation, and have non-biologically plausible architectures (Cox, 2024).

★ This leads us to the question...

How much can we get out of a Hopfield-like network using Hebbian learning?

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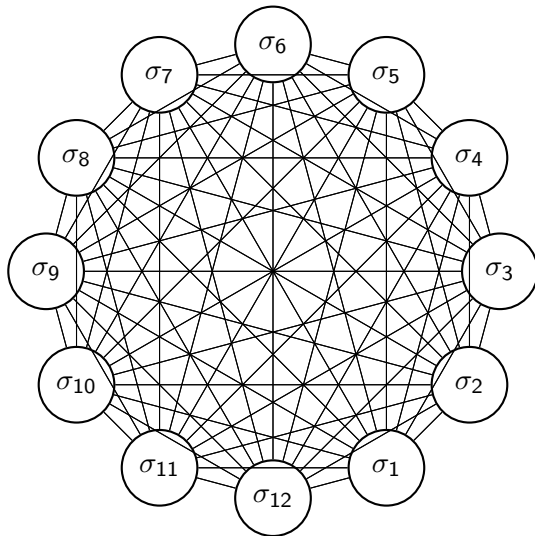
★ Hopfield Networks

Definition (Hopfield Networks)

A **Hopfield network** (J. Hopfield, 1982) is a single-layer neural network of D binary neurons, (typically from the set $\{-1, 1\}$) with full lateral connections (no self-connections). The weights between each neuron and the others is the $D \times D$ symmetric matrix W , with the diagonal all zeros.

This network architecture is simple, and can be reformulated as a feedforward network: but what do we even do with it?

Hopfield Network Architecture



Fully connected network with $D = 12$ neurons

Hopfield recall

Definition (Hopfield energy; pattern matrix)

The **energy** of a Hopfield network is a function of the current state of the neurons in the Hopfield network at a time t , and the stored patterns in the matrix at time t , which are the rows of the $N \times D$ **pattern matrix** $\Xi = [\xi^1, \xi^2, \dots, \xi^N]$:

$$E_{\Xi}(\sigma^t) = -\frac{1}{2} \sum_{\mu=1}^N \left(\sum_{i=1}^D \xi_i^{\mu} \sigma_i^t \right)$$

The intuition here is that the energy describes how similar the current state of the network is to the desired patterns. Recall is **energy minimization**, where, ideally, local energy minima correspond with stored patterns.

★ Hopfield recall

Definition (Asynchronous and synchronous Hopfield recall)

Given an initial Hopfield network state $\sigma^{(t=0)}$, recall is either performed asynchronously or synchronously. The asynchronous definition updates each index of the state at random such that:

$$\sigma_i^{(t+1)} = \text{sgn} [E(\sigma_{i-}^t) - E(\sigma_{i+}^t)],$$

where $\sigma_{i\pm}^t$ is the state vector σ^t with the i 'th set to $+1$ or -1 . This can be reformulated synchronously (Krotov et al., 2025), as:

$$\sigma^{(t=T)} = \text{sgn} [W\sigma^{t=0}],$$

with T being the final state of the network.

★ Alternative Formulation

Hopfield recall (reformulated)

Let us have $N \times D$ pattern matrix $\Xi = [\xi^1, \xi^2, \dots, \xi^N]$. Then, synchronous Hopfield recall is:

$$\sigma^{(t=T)} = \text{sgn} \left[\xi^1 (\xi^1)^\top \sigma^{(0)} + \dots + \xi^N (\xi^N)^\top \sigma^{(0)} \right]$$

While recall and training are **distinct** modes of the network, this alternative formulation makes clear what using the **Activity Product Rule** for updating weights does; i.e. $\Delta W = \xi^\mu (\xi^\mu)^\top$.

★ Dense Associative Memory

Definition

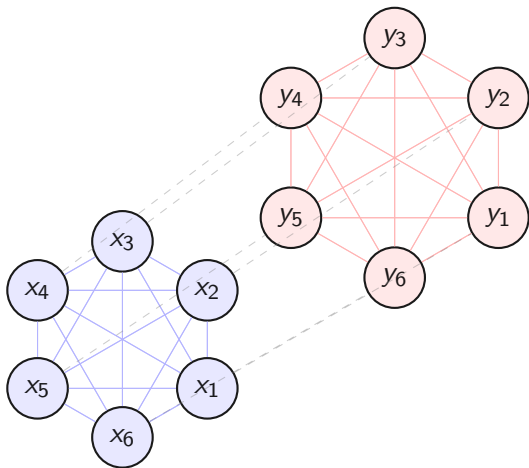
Dense Associative Memories generalize Hopfield networks (Demircigil et al., 2017; Krotov & Hopfield, 2016). Let us have an $N \times D$ pattern matrix $\Xi = [\xi^1, \xi^2, \dots, \xi^N]$, where each ξ^μ , $\mu = 1, 2, \dots, N$ is a pattern we wish to store. Let F_n be a **polynomial transmission function** of the form:

$$F_n(x) = \frac{1}{n}(x)^n.$$

Then, the energy of the Dense Associative Memory is:

$$E(\sigma^t) = - \sum_{\mu=1}^N F_n \left(\sum_{i=1}^D \xi_i^\mu \sigma_i \right).$$

Dense Associative Memory Architecture



The Memory Tesseract (Kelly et al., 2017).

Dense Associative Memory recall

Definition (DAM recall)

Asynchronously, we update the state vector in the same way as the Hopfield network (Def. 3). But, with the polynomial transmission function F_n so defined, can be put in terms of the derivative of F_n (Krotov et al., 2025):

$$\sigma^{(t=T)} = \text{sgn} \left[\sum_{\mu=1}^N \xi^{\mu} F'_n \left(\sum_{i=1}^D \xi_i^{\mu} \sigma_i^{(t=0)} \right) \right].$$

Examples of DAM generalization

Example

Hopfield networks are Dense Associative Memories with polynomial transmission function F_2 .

Example

Minerva2 models (Hintzman, 1984) are roughly Dense Associative Networks, assuming that all patterns $\xi^1, \xi^2, \dots, \xi^N$ and query patterns are normalized, and with polynomial transmission function F_4 .

★ Capacity Proofs

DAM Capacity

The **critical capacity** C of a Dense Associative Memory is the number of patterns that the network can store and reasonably recall with minimal error. The maximum capacity C is a function of the dimension of the state vector, to the power of $n - 1$ from F_n (Bao et al., 2022; Demircigil et al., 2017; Krotov & Hopfield, 2016):

$$C = \alpha D^{n-1},$$

where α is some constant.

Capacities

Example

The capacity of a Hopfield network $C \approx 0.14 * D$ (Amit et al., 1987; J. Hopfield, 1982).

Example

The capacity of Minerva2 is $C = \alpha D^3$.

Hebbian Learning

- (a) Dense Associative Memories have been shown to be **equivalent to multi-head attention** in the transformer architecture (Ramsauer et al., 2021; Vaswani et al., 2023)
- (b) But they achieve this power at the cost of their Hebbian plausibility
- (c) Hopfield networks are usually trained using the Activity Product Rule (Haykin, 2009)

Definition (Activity Product Rule)

The **Activity Product Rule** updates the weights W of a network by the outer product of the pre- and post-synaptic layers x_i and x_j (respectively), multiplied by some learning rate η :

$$\Delta W = \eta x_i x_j^T.$$

★ General Hebbian update rule

Definition (General Hebbian Form)

A **Hebbian** learning rule for pre-synaptic layer x_i and post-synaptic layer x_j , with weights W is of the form:

$$\Delta W_{ij} = f(W_{ij}; x_i, x_j).$$

This definition contains an incredibly large class of functions: what is an interesting subset?

Definition (Gerstner and Kistler (2002))

If one expands $\Delta W_{ij} = f(W_{ij}; x_i, x_j)$ about $(0,0)$, we get:

$$\begin{aligned}\Delta W_{ij} = f(W_{ij}; x_i, x_j) &= c_0(W_{ij}) + c_1^{\text{pre}}(W_{ij})x_j + c_1^{\text{post}}x_i \\ &+ c_2^{\text{pre}}(W_{ij})x_j^2 + c_2^{\text{post}}(W_{ij})x_i^2 + c_2^{\text{corr}}(W_{ij})x_ix_j \\ &+ \mathcal{O}(x),\end{aligned}$$

which captures many interesting Hebbian learning rules for binary patterns.

Example

The Activity Product Rule is just the expanded Hebbian learning rule with c_2^{corr} set to some constant, and all other constants set to 0.

Example

Oja's Rule (Oja, 1982) by setting $c_2^{\text{corr}} = \eta_0 > 0$, and $c_2^{\text{post}} = -\eta_0 w_{ij}$, with all other coefficients set to 0.

☆ Project Proposal

Binary Hebbian Associative Memories

Consider a network $\mathcal{M}$ with units $\sigma_i \in \{-1, 1\}$, $i = 1, 2, \dots, D$, and weight $D \times D$ weight matrix:

$$W_{ij}^{(t+1)} = \begin{cases} f(W_{ij}^{(t)}; \xi_i^\mu, \xi_j^\mu), & \text{if } i \neq j, \\ 0, & \text{otherwise,} \end{cases}$$

with Hebbian learning rule f of the form described in Definition 12. Assume that the synchronous recall of $\mathcal{M}$ is just:

$$\mathcal{M}(\sigma) = \text{sgn}[W\sigma].$$

What is the maximum capacity C of patterns that $\mathcal{M}$ can reliably store and retrieve?

Hebbian Network Recall...

With W formulated as above: this makes synchronous recall

$$\mathcal{M}(\sigma) = \text{sgn} \left[f(W^{(1)}; \xi^1, \xi^1)(\sigma) + \cdots + f(W^{(t=\mu)}; \xi^\mu, \xi^\mu)(\sigma) \right],$$

for $\mu = 1, \dots, N$.

☆ How can we solve this problem

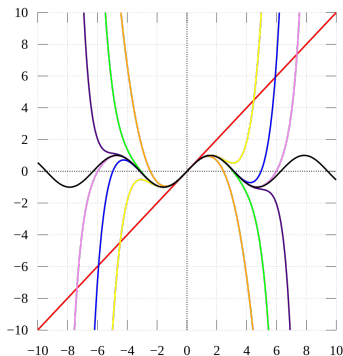


Figure: Taylor series approximation of $\sin(x)$ (IkamusumeFan, 2013).

- (a) Demonstrating, in general, how the capacity C of $\mathcal{M}$ scales as a function of the dimension of the network D and the coefficients

☆ Theoretical Approach

- (a) We assume the definition of $\mathcal{M}$ above, and then provide a general function for how capacity grows
- (b) No need for proof to be too specific (e.g., Amit et al. (1987), Bao et al. (2022), and Robert J. McEliece et al. (1987))
- (c) Rather, proof strategy is more informal: Demircigil et al. (2017) and Krotov et al. (2025)

Proof Strategy

Assume binary patterns $\xi^1, \xi^2, \dots, \xi^N$ with each element sampled indifferently from $\{-1, 1\}$. Train a network $\mathcal{M}$ with the function f . Assume we are trying to recall ξ^1 . **What is the maximum number N where the noise overtakes true signal in recall?**

Signal-to-noise

Overview of Krotov et al. (2025) applied to $\mathcal{M}$:

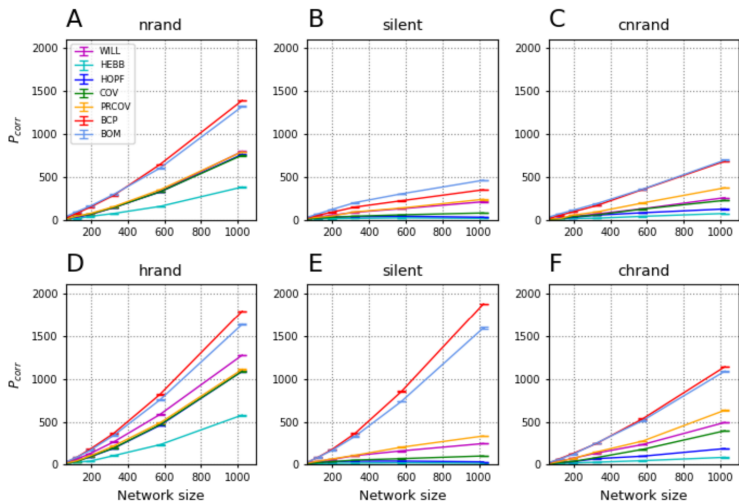
$$\begin{aligned}\mathcal{M}(\sigma) &= \operatorname{sgn} \left[\sum_{\mu=1}^N f(W^{\mu}; \xi^{\mu}, \xi^{\mu})(\sigma) \right] \\ &= \operatorname{sgn} \left[\underbrace{f(W^1; \xi^1, \xi^1)(\sigma)}_{\text{signal}} + \cdots + \underbrace{f(W^N; \xi^N, \xi^N)(\sigma)}_{\text{noise}} \right].\end{aligned}$$

☆ Experimental Approach

Lansner et al. (2025)

An interesting subset of Hebbian learning rules is analyzed in Lansner et al. (2025), **showing that some Hebbian learning rules have low-power polynomial capacity scaling.**

Capacity was estimated by fitting a curve to experimental evidence.



☆ Generating Hebbian Networks

- (a) Treat coefficients as hyperparameters, and perform hyperparameter search
- (b) Measure best performance on a collection of tasks
- (c) Large-scale cognitive models: (Jamieson et al., 2018; Johns & Jones, 2015; Ozen et al., 2022)

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